Unit

7 Ionic and metallic bonds

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Ionic bonds
- Names and chemical formulae of ionic compounds
- Metallic bonds in metals

7.1 Conductors, electrolytes and non-conductors

We can classify substances into three groups according to how they conduct electricity.

- **Conductors** These are substances which conduct electricity but are not chemically changed during conduction. For example, metals are conductors (Fig. 7.1).
- **Electrolytes** These are compounds which conduct electricity in *molten* state or *aqueous* solution. They are decomposed by electricity during conduction. Compounds made up of metals and non-metals are electrolytes (Fig. 7.2).
- **Non-conductors** These are substances which do not conduct electricity in solid, molten state or aqueous solution. All non-metals (except graphite) are non-conductors. Compounds made up of non-metals are also non-conductors◂ (Fig. 7.3).

DSE 2012 Paper 1A Q1

A few compounds made up of non-metals are electrolytes which conduct electricity in aqueous solution. Examples are hydrogen chloride, sulphur dioxide, etc.

7.1

Finding out which compounds can conduct electricity.

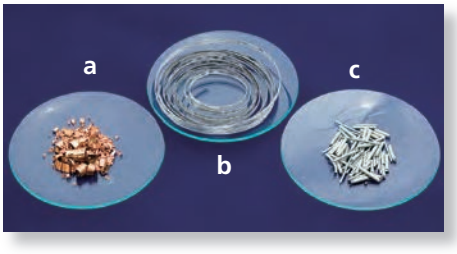


Fig. 7.1 Some conductors: (a) copper (b) magnesium (c) iron



Fig. 7.2 Some electrolytes: (a) sodium chloride (b) lead(II) bromide (c) potassium iodide



Fig. 7.3 Some non-conductors: (a) sugar (b) ethanol (c) distilled water

conductor 導電體　electrolyte 電解質　molten 熔融的　aqueous 水的　non-conductor 非導電體

Fig. 7.4 summarizes the electrical conductivities of different substances.

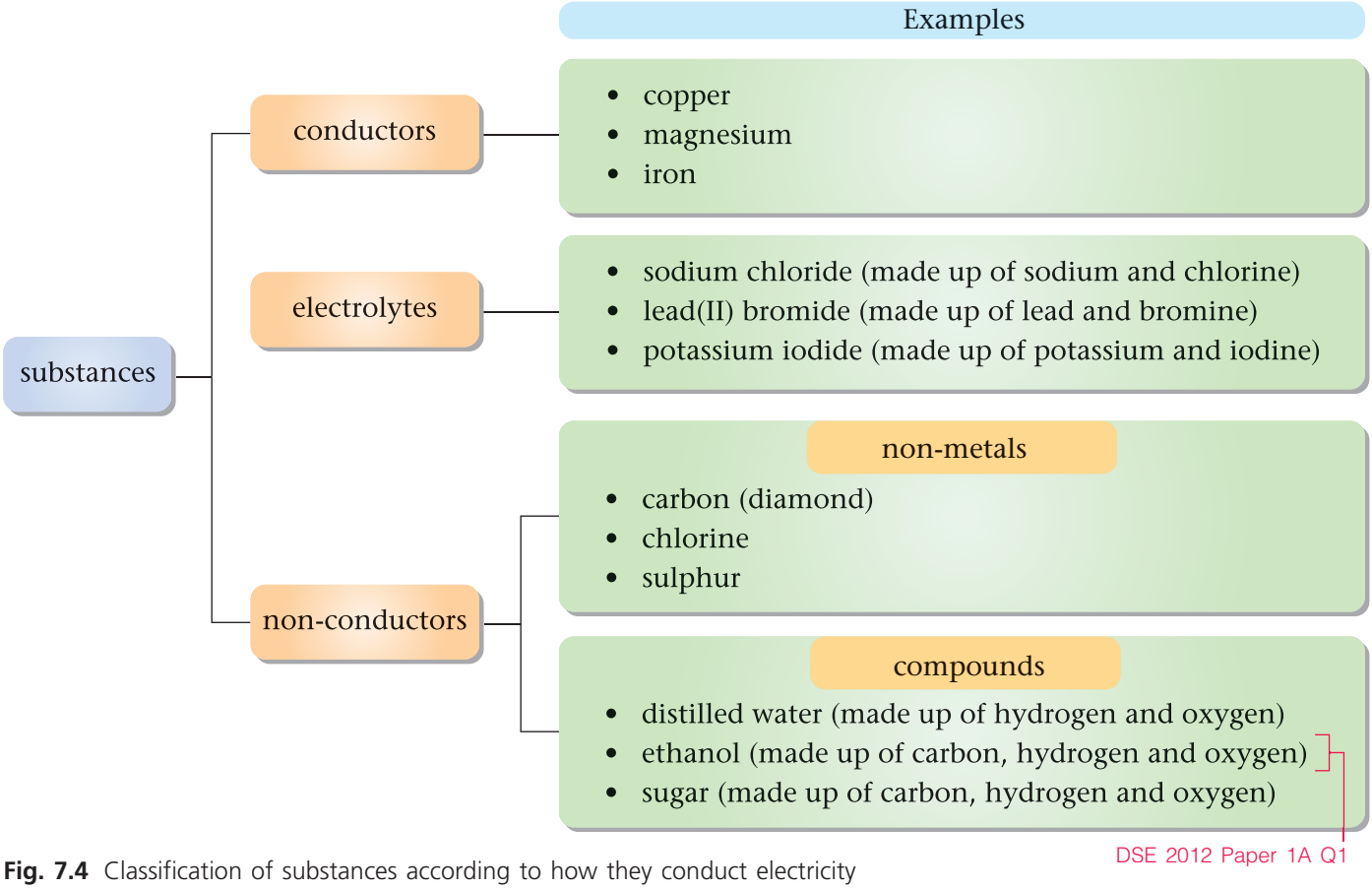


Fig. 7.4 Classification of substances according to how they conduct electricity

7.2 Evidence of ions from electrolysis of molten lead(II) bromide

DSE 2012 Paper 1A Q1

Lead(II) bromide is harmful.

Lead(II) bromide is an electrolyte. Let us look at what happens when we pass electricity through molten lead(II) bromide[◀] (Fig. 7.5). N1

When we pass electricity through molten lead(II) bromide, a reddish brown gas (bromine) is formed at the positive electrode. A white shiny solid (lead) is formed at the negative electrode. Lead(II) bromide is decomposed into lead and bromine by electricity. N2

$$\text{lead(II) bromide} \xrightarrow{\text{electricity}} \text{lead} + \text{bromine}$$

N1 When heating is stopped and the molten lead(II) bromide is allowed to cool down, the light bulb gradually goes out. At lower temperatures, movement of ions in the molten lead(II) bromide slows down and a smaller current flows through the external circuit. No current flows when the molten lead(II) bromide becomes solid.

N2 Bromine vapour will be liberated in the process, NOT liquid bromine or bromine solution.

The observations provide evidence for the existence of positive lead(II) ions (Pb^{2+}) and negative bromide ions (Br^-) in lead(II) bromide.

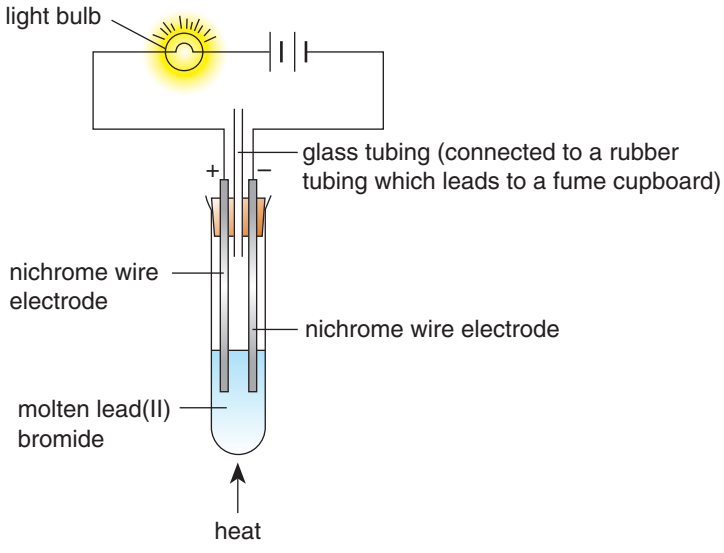


Fig. 7.5 Passing electricity through molten lead(II) bromide

In solid state, ions in the compound are held together by strong attraction. They are not free to move. Hence solid lead(II) bromide does not conduct electricity.

When lead(II) bromide becomes molten, the lead(II) ions (Pb^{2+}) and bromide ions (Br^-) become mobile.

lead(II) bromide ⟶ lead(II) ions + bromide ions

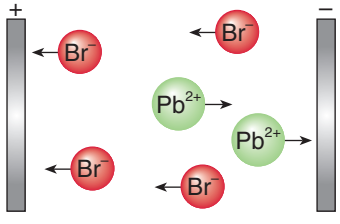


Fig. 7.6 Bromide ions and lead(II) ions move towards the positive electrode and negative electrode respectively

The negative bromide ions move towards the positive electrode (Fig. 7.6). These ions give up electrons to the electrode and become atoms. The atoms then join in pairs to form molecules.

bromide ions – electrons ⟶ bromine atoms
⟶ bromine molecules

The positive lead(II) ions move towards the negative electrode (Fig. 7.6). These ions receive electrons from the electrode and become atoms.

lead(II) ions + electrons ⟶ lead atoms

Compounds made up of metals and non-metals are electrolytes. These compounds consist of positive and negative ions.

7.3 Chemical bonds

We know that the electronic arrangements of atoms of noble gases are extremely stable. Atoms of other elements tend to undergo changes to obtain such a stable electronic arrangement. They do this by joining or bonding together. They may bond with atoms of the same type (e.g. a hydrogen atom bonds with another hydrogen atom) or with atoms of a different type (e.g. a hydrogen atom bonds with a chlorine atom).

The forces holding the particles together are called **chemical bonds**. The properties of substances are affected by the type of chemical bonding between the particles.

7.4 Ionic bonds

Reference Website
Refer to a video clip on ionic bonding:
http://www.bbc.co.uk/schools/gcsebitesize/science/add_ocr_gateway/periodic_table/ionicact.shtml
(accessed July 2014)

Atoms of metals have one, two or three outermost shell electrons. So, the easy way for them to obtain the electronic arrangement of an atom of a noble gas is to lose their outermost shell electrons.

Atoms of non-metals have spaces in their outermost shells. So, they need to gain electrons in order to obtain the electronic arrangement of an atom of a noble gas.

A metal and a non-metal can react together because the electrons lost by atoms of the metal can be taken up by atoms of the non-metal. This transfer of electrons results in an **ionic bond**.

Ionic bond in sodium chloride

A sodium atom has an electronic arrangement of 2,8,1. It tends to lose one electron to obtain the electronic arrangement of a neon atom (2,8). A sodium ion (Na^+) is formed when a sodium atom loses one electron (Fig. 7.7).

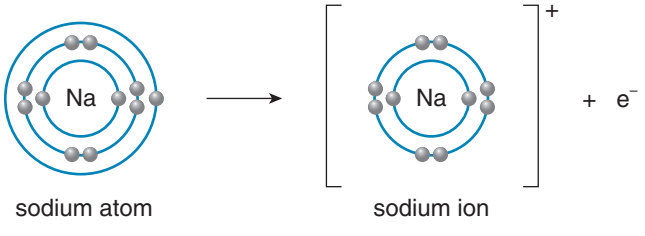


Fig. 7.7 Formation of a sodium ion

Animation
Ionic bond

chemical bond 化學鍵　　ionic bond 離子鍵

A chlorine atom has an electronic arrangement of 2,8,7. It needs to gain one electron in order to obtain the electronic arrangement of an argon atom (2,8,8). A chloride ion (Cl^-) is formed when a chlorine atom gains one electron (Fig. 7.8).

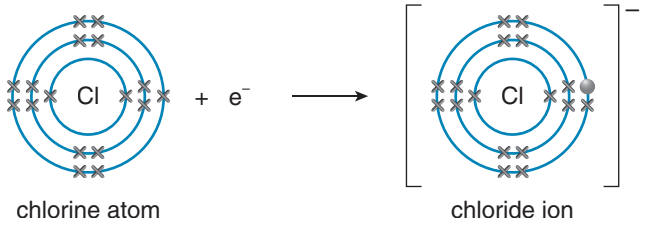
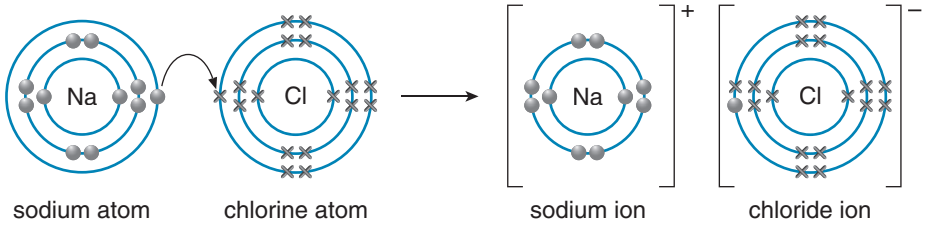


Fig. 7.8 Formation of a chloride ion

N3 Remind students that only electrons, but not protons and neutrons, of atoms are affected in chemical reactions. The protons and neutrons will only be affected in nuclear reactions.

Atoms of sodium and chlorine can each obtain a stable electronic arrangement by the transfer of one electron from the sodium atom to the chlorine atom. In the process of electron transfer, each atom becomes an ion with an electronic arrangement of an atom of a noble gas (Fig. 7.9). N3

N4 Examination questions often show the electron diagram of a compound and ask students to identify the elements that make up the compound.



N4

Fig. 7.9 Electron transfer during the reaction between sodium and chlorine



Fig. 7.10 Sodium chloride is a compound of the dangerously reactive elements sodium and chlorine

There are strong **electrostatic forces of attraction** between positive and negative ions. This attraction, which holds the ions together, is a chemical bond called an ionic bond. The compound produced, sodium chloride (Fig. 7.10), in this case, is an **ionic compound**.

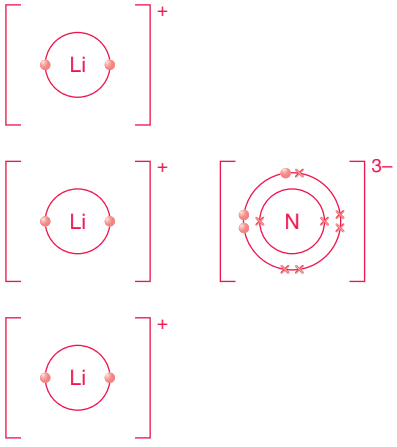
Any Group I metal reacts with any Group VII non-metal in the same way. For example, potassium and bromine react to form potassium bromide, which is made from potassium ions and bromide ions.

✓ An ionic bond is the strong electrostatic forces of attraction between oppositely charged ions.

✓ An ionic bond is formed when one or more electrons are transferred from one atom (or group of atoms) to another.

electrostatic force of attraction 静電引力　　ionic compound 離子化合物

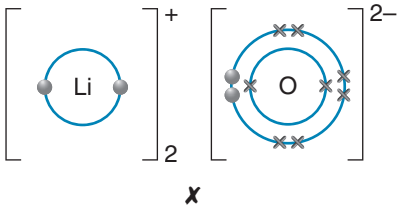
N5 Lithium reacts with nitrogen to form lithium nitride (Li_3N).



N6 A piece of freshly cut lithium tarnishes after exposed to air for some time. Lithium oxide is formed.

N7 Caesium (Cs) is a Group I element. It reacts with oxygen to form an oxide with the chemical formula Cs_2O.

It is WRONG to draw the electron diagram of lithium oxide in the following way:



✓ When a metal and a non-metal combine to form an ionic compound, the metallic atoms release electrons while the non-metallic atoms gain electrons.

Ionic bond in lithium oxide

Consider the reaction between lithium and oxygen. A lithium atom has an electronic arrangement of 2,1. It tends to lose one electron to obtain the electronic arrangement of a helium atom (2). N5

An oxygen atom has an electronic arrangement of 2,6. It needs to gain two electrons in order to obtain the electronic arrangement of a neon atom (2,8).

When lithium reacts with oxygen to form lithium oxide, electrons from every two lithium atoms, one from each, are transferred to one oxygen atom. Lithium ions and oxide ions are formed. Ionic bonding holds the lithium ions and oxide ions together (Fig. 7.11). N6 N7

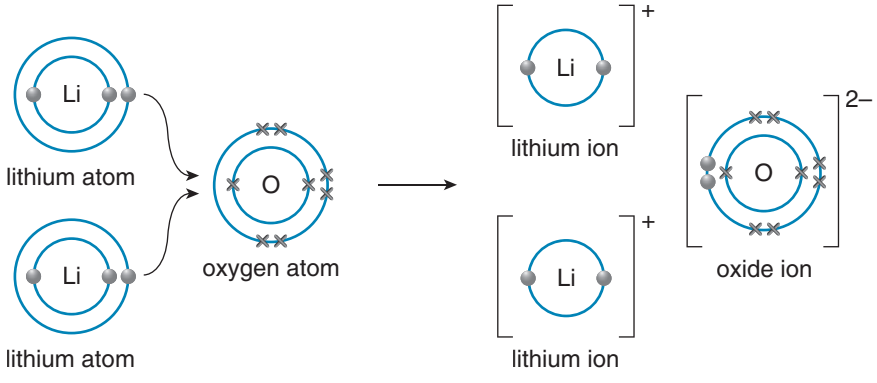


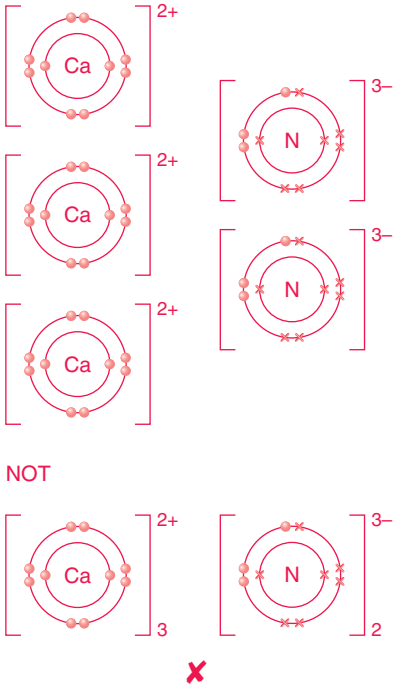
Fig. 7.11 Electron transfer during the reaction between lithium and oxygen

Ionic bond in magnesium nitride

Consider the reaction between magnesium and nitrogen. A magnesium atom has an electronic arrangement of 2,8,2. It tends to lose two electrons to obtain the electronic arrangement of a neon atom (2,8).

A nitrogen atom has an electronic arrangement of 2,5. It needs to gain three electrons in order to obtain the electronic arrangement of a neon atom (2,8).

N8 Calcium reacts with nitrogen to form calcium nitride (Ca_3N_2). Examination questions often ask about the chemical formula and electron diagram of calcium nitride.



Simulation
Ionic bond formation

When magnesium reacts with nitrogen to form magnesium nitride, electrons from every three magnesium atoms, two from each, are transferred to two nitrogen atoms. Magnesium ions and nitride ions are formed. Ionic bonding holds the magnesium ions and nitride ions together (Fig. 7.12).

CE 2009 Paper 1 Q2(a)(ii)(2)

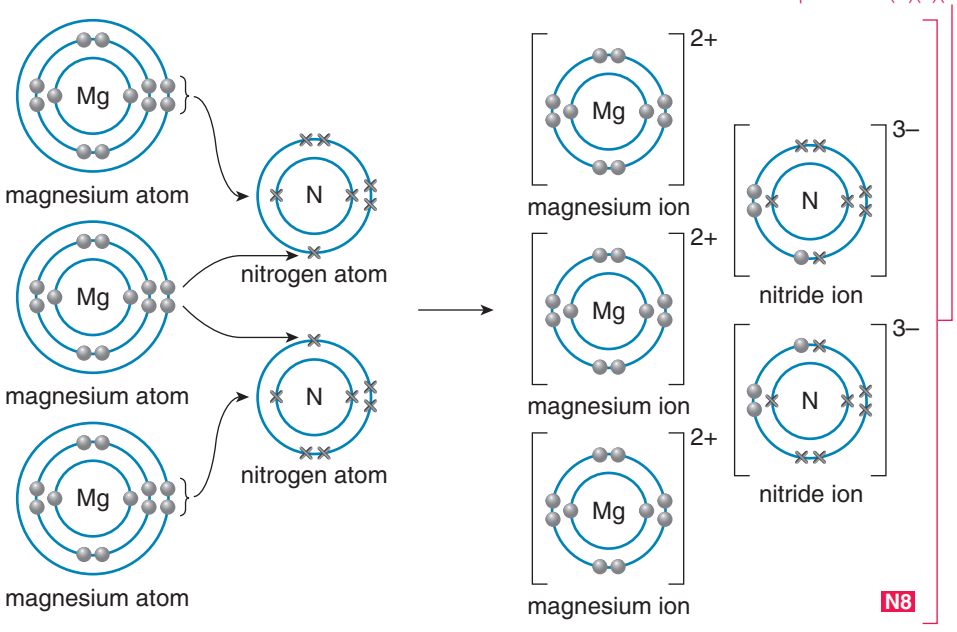
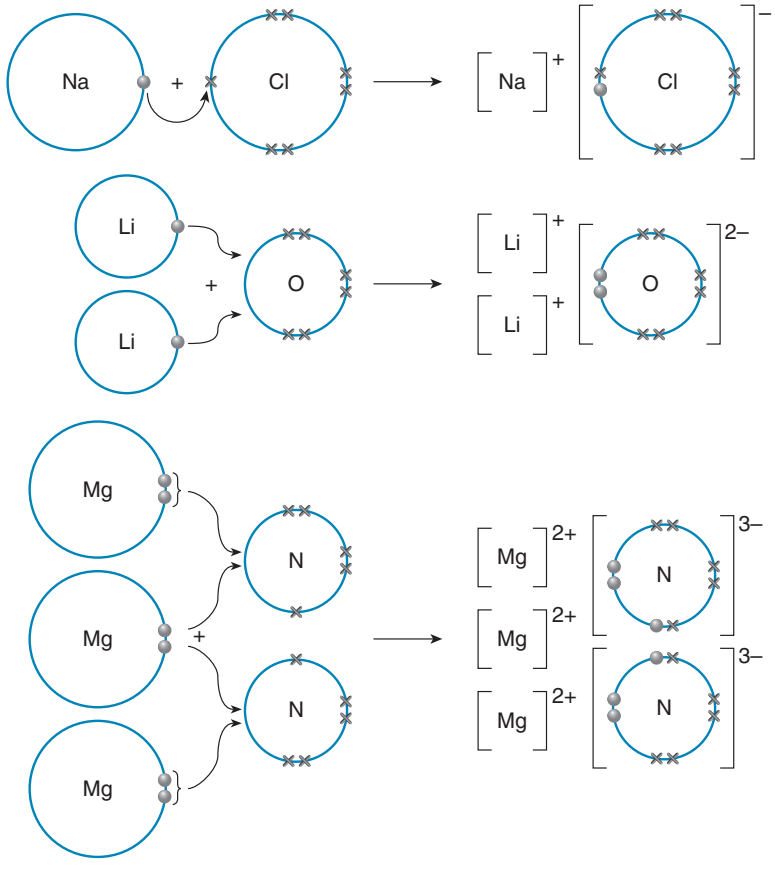


Fig. 7.12 Electron transfer during the reaction between magnesium and nitrogen

Since only the outermost shell electrons are involved in electron transfer, we can simplify Figs. 7.9, 7.11 and 7.12 as below:



Example 7.1

Q X, Y and Z are three different elements. X^{2+} ion, Y^- ion and argon atom have the same electronic arrangement.

a) What are elements X and Y?

b) Draw an electron diagram of the compound formed from elements X and Y, showing electrons in the *outermost shells* only.

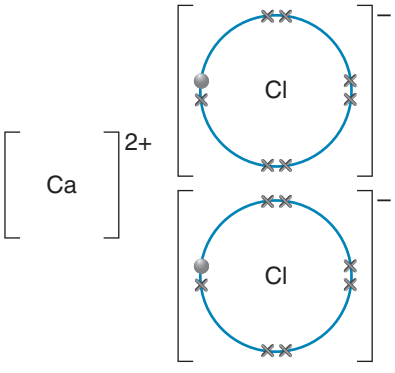
c) Element Z belongs to the same period as element X and the same group as element Y in the periodic table. What is element Z?

A

a) Element X is calcium.

Element Y is chlorine.

b)



c) Bromine

Practice 7.1

1 Use an electron diagram to show the electron transfer when each of the following pairs of elements react.
Refer to page T29.

a) magnesium and oxygen

b) sodium and nitrogen

2 The following table lists the atomic numbers of four elements.

Element	W	X	Y	Z
Atomic number	2	9	10	13

a) Which TWO elements would form an ionic compound? X and Z

b) Draw an electron diagram of the compound formed, showing electrons in the *outermost shells* only.
Refer to page T29.

7.5 Compounds containing polyatomic ions

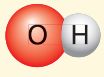
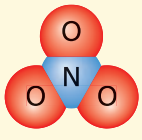
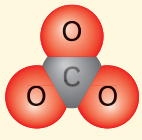
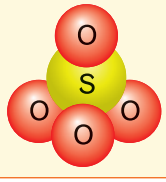
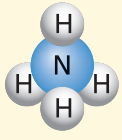
Up to now we have been looking at simple ions that are formed from a single atom only. Actually an ion can also be formed from a group of atoms. This is called a **polyatomic ion**.

For example, carbonate ion (CO_3^{2-}) is a polyatomic ion. It is formed from one carbon atom and three oxygen atoms. The whole group of four atoms carries 2 negative charges.

Table 7.1 shows some examples of polyatomic ions.

Table 7.1

Examples of polyatomic ions

Name	Hydroxide	Nitrate	Carbonate	Sulphate	Ammonium
Chemical formula	OH^-	NO_3^-	CO_3^{2-}	SO_4^{2-}	NH_4^+
Model					

N9 Students should memorize the charges and formulae of the common ions as these information are essential for writing the chemical formulae of ionic compounds. Such knowledge is also essential for other topics of the course.

7.6 Names of ions N9

Names of positive ions

Table 7.2 lists the names of some common positive ions. If a metal forms only one kind of positive ion, the name of the ion is the same as the metal. For example, potassium (K) forms potassium ion (K^+).

H^+ and NH_4^+ are positive ions formed from non-metals.

Table 7.2

Names of some common positive ions

With 1 positive charge		With 2 positive charges		With 3 positive charges	
Chemical formula	Name	Chemical formula	Name	Chemical formula	Name
Li^+	lithium ion	Mg^{2+}	magnesium ion	Al^{3+}	aluminium ion
Na^+	sodium ion	Ca^{2+}	calcium ion	Fe^{3+}	iron(III) ion
K^+	potassium ion	Zn^{2+}	zinc ion		
Ag^+	silver ion	Fe^{2+}	iron(II) ion		
H^+	hydrogen ion	Cu^{2+}	copper(II) ion		
NH_4^+	ammonium ion	Pb^{2+}	lead(II) ion		
Cu^+	copper(I) ion				

polyatomic ion 多原子離子

Some metals can form more than one kind of positive ion. For example, copper can form two kinds of positive ions, one carrying 1 positive charge and one carrying 2 positive charges (Cu^+ and Cu^{2+}). When naming these ions, write a Roman numeral in brackets after the name of the metal to show the number of positive charges. Thus, we use the name of copper(I) ion for Cu^+, and copper(II) ion for Cu^{2+} (Table 7.3).

Table 7.3

Examples of metals that can form more than one kind of positive ion

Ion	Name of ion	Example of compound
Cu^+	copper(I) ion	copper(I) oxide
Cu^{2+}	copper(II) ion	copper(II) oxide
Fe^{2+}	iron(II) ion	iron(II) chloride
Fe^{3+}	iron(III) ion	iron(III) chloride

Fig. 7.13 and Fig. 7.14 show oxides of copper and chlorides of iron respectively.



Fig. 7.13 Copper(I) oxide (left) and copper(II) oxide (right)



Fig. 7.14 Iron(II) chloride (left) and iron(III) chloride (right)

Names of negative ions

Negative ions include all simple ions formed from non-metals (except H^+) and most polyatomic ions (for example, SO_3^{2-} and CO_3^{2-}).

Simple negative ions have names ending in '-ide'. Polyatomic ions containing oxygen have names ending in -ite or -ate. The polyatomic ion with less oxygen is named -ite, and that with more oxygen is named -ate.

A polyatomic ion formed from an oxygen atom and a hydroxide ion (OH^-). It is an exception.

For example,

NO_2^-	nitr**ite**	and	NO_3^-	nitr**ate**
SO_3^{2-}	sulph**ite**	and	SO_4^{2-}	sulph**ate**

Table 7.4 lists the names of some common negative ions.

Table 7.4

Names of some common negative ions

With 1 negative charge		With 2 negative charges		With 3 negative charges	
Chemical formula	**Name**	**Chemical formula**	**Name**	**Chemical formula**	**Name**
F^-	fluoride ion	O^{2-}	oxide ion	N^{3-}	nitride ion
Cl^-	chloride ion	S^{2-}	sulphide ion	PO_4^{3-}	phosphate ion
Br^-	bromide ion	SO_3^{2-}	sulphite ion		
I^-	iodide ion	SO_4^{2-}	sulphate ion		
NO_2^-	nitrite ion	CO_3^{2-}	carbonate ion		
NO_3^-	nitrate ion	$Cr_2O_7^{2-}$	dichromate ion		
OH^-	hydroxide ion				
HCO_3^-	hydrogencarbonate ion				
HSO_4^-	hydrogensulphate ion				
MnO_4^-	permanganate ion				

7.7 Naming ionic compounds

When naming an ionic compound, name the positive ion first, followed by the negative ion. For example, a compound which consists of sodium ions and chloride ions is named sodium chloride. Table 7.5 gives further examples.

Table 7.5

Names of some ionic compounds

Positive ion in the compound	Negative ion in the compound	Name of the compound
lithium ion	oxide ion	lithium oxide
magnesium ion	nitride ion	magnesium nitride
ammonium ion	chloride ion	ammonium chloride
calcium ion	hydroxide ion	calcium hydroxide
copper(II) ion	carbonate ion	copper(II) carbonate
iron(III) ion	nitrate ion	iron(III) nitrate
iron(II) ion	sulphate ion	iron(II) sulphate
potassium ion	permanganate ion	potassium permanganate

7.2
Examining colours of aqueous solutions of ionic compounds.

7.8 Colours of ionic compounds

If an ionic compound has colour, the colour may arise from either the negative or positive ion, or even from both ions.

Consider the colours of the aqueous solutions of potassium chloride and potassium dichromate. The aqueous solution of potassium chloride is colourless (Fig. 7.15). Hence the potassium ions must be colourless. Since the aqueous solution of potassium dichromate is orange in colour (Fig. 7.16), the orange colour must come from the dichromate ions.



Fig. 7.15 Colourless potassium chloride solution



Fig. 7.16 Orange potassium dichromate solution

N10 Examination questions may ask students to deduce the colours of ions from the colours of aqueous solutions of some ionic compounds.
e.g.

Compound	Colour of aqueous solution
QT	green
XZ	yellow
XT	colourless

Answer:
$Q^{2+}(aq)$ — green
$X^{2+}(aq)$ — colourless
$T^{2-}(aq)$ — colourless
$Z^{2-}(aq)$ — yellow

Table 7.6 and Fig. 7.17 show the colours of some ions in aqueous solutions. N10

Table 7.6

Colours of some ions in aqueous solutions

Ion	Chemical formula	Colour
Iron(II)	Fe^{2+}	pale green
Iron(III)	Fe^{3+}	yellow-brown
Copper(II)	Cu^{2+}	blue or green
Permanganate	MnO_4^-	purple
Dichromate	$Cr_2O_7^{2-}$	orange
Chromium(III)	Cr^{3+}	green
Nickel(II)	Ni^{2+}	green
Manganese(II)	Mn^{2+}	very pale pink (or colourless)

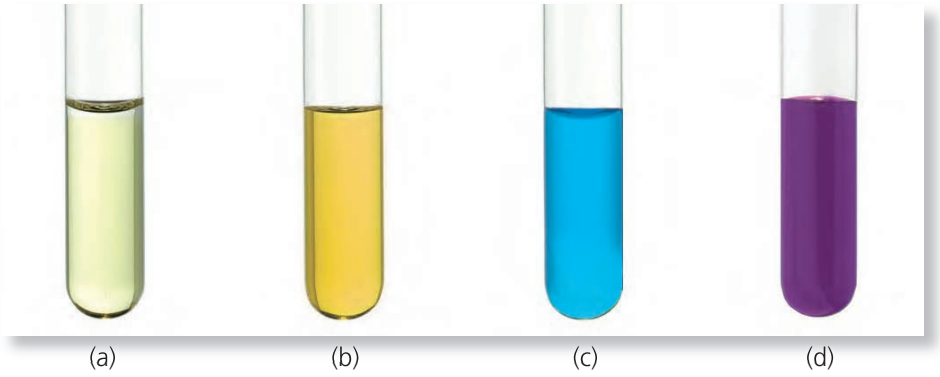


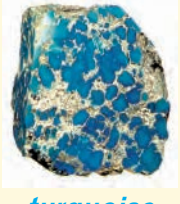
Fig. 7.17 Aqueous solutions containing (a) iron(II) ions (b) iron(III) ions (c) copper(II) ions (d) permanganate ions

Colours of gemstones

The colours of gemstones are due to the presence of traces of coloured ions. Some examples are listed in Table 7.7.

Table 7.7

Coloured ions in gemstones

Gemstone	Colour	Ion present	Chemical formula of ion
 *jade*	green	chromium(III)	Cr^{3+}
 *peridot*	light green	iron(II)	Fe^{2+}
 *topaz*	yellow-brown	iron(III)	Fe^{3+}
 *turquoise*	greenish blue	copper(II)	Cu^{2+}

jade 翡翠 peridot 橄欖石 topaz 黃玉 turquoise 綠松石

Movement of coloured ions

7.3
Observing the migration of ions.

Video
Observing the migration of ions

We can observe the movement of ions using the set-up shown in Fig. 7.18. When we place a small crystal of potassium permanganate at the centre of a strip of filter paper moistened with tap water, we can see a purple spot slowly moving towards the positive electrode. This is because negative permanganate ions which are purple in colour move towards the positive electrode. Positive potassium ions move towards the negative electrode. However, we cannot see the potassium ions because they are colourless.

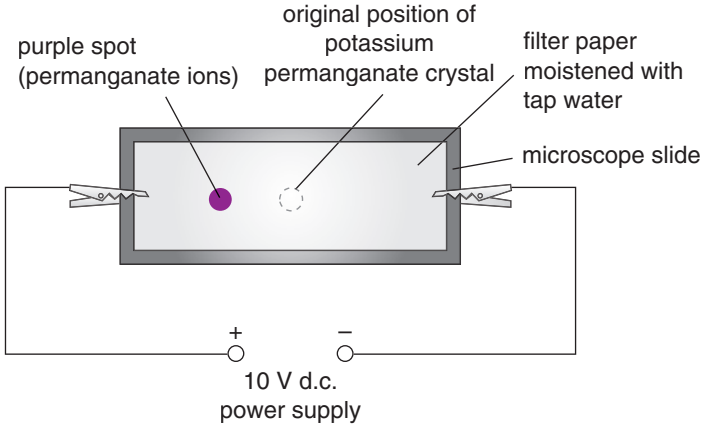


Fig. 7.18 Movement of permanganate ions when electricity is passed through

Hydrogen ions (H^+) around the negative electrode will gain electrons to form hydrogen.

Consider the set-up shown in Fig. 7.19. Electricity is passed through a gel containing copper(II) ions and dichromate ions. An orange colour appears near the positive electrode. This is because negative dichromate ions move towards the positive electrode. A blue colour appears near the negative electrode. This is because positive copper(II) ions move towards the negative electrode.

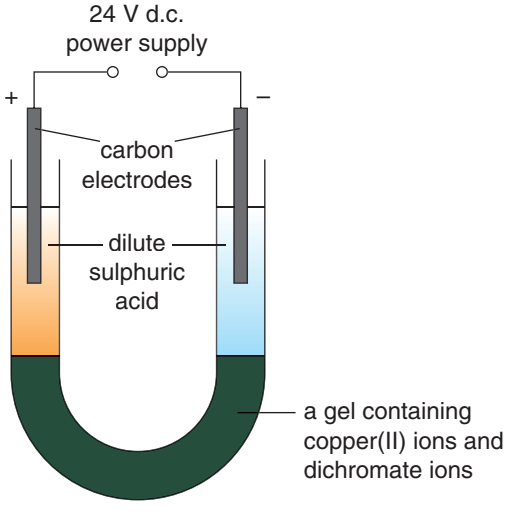


Fig. 7.19 Movement of ions when electricity is passed through a gel containing copper(II) ions and dichromate ions N11

N11 This experiment can be used to show that ions move towards oppositely charged electrodes.

Practice 7.2

1 Topaz is yellow-brown in colour. Suggest the ion responsible for the colour.
Iron(III) ions

2 The table below lists the colours of the aqueous solutions of three ionic compounds.

Compound	Colour of aqueous solution
QT	blue
XZ	yellow
XT	colourless

Deduce the colours of the following ions:

a) $Q^{2+}(aq)$ ion; blue

b) $X^{2+}(aq)$ ion; colourless

c) $T^{2-}(aq)$ ion; colourless

d) $Z^{2-}(aq)$ ion. yellow

CE 2008 Paper 2 Q9

7.9 Chemical formulae of ionic compounds

Game
Chemical formulae killer

A **chemical formula** is a way of representing a chemical substance using symbols and figures.

The chemical formula of an ionic compound shows:

- the types of ions present; and
- the ratio of one type of ion to the other.

Compounds are electrically neutral. Thus, in an ionic compound, there must be the right number of each type of ion so that the total positive charges equal the total negative charges.

Writing chemical formulae of ionic compounds

We write the chemical formula of an ionic compound by combining the symbols of its positive and negative ions. Let us work out the chemical formulae of sodium chloride, lithium oxide, magnesium nitride and calcium hydroxide.

In writing the chemical formula of an ionic compound, write the symbol of the positive ion first, followed by the symbol of the negative ion.

chemical formula 化學式

1 **Sodium chloride** The sodium ion (Na^+) carries 1 positive charge while the chloride ion (Cl^-) carries 1 negative charge. For electrical neutrality, we need one Cl^- ion for every Na^+ ion. So the chemical formula of sodium chloride is NaCl (Fig. 7.20).

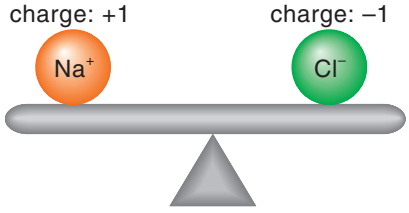


Fig. 7.20 Writing the chemical formula of sodium chloride

N12 There is no need to use brackets for a simple ion, even when the number of that ion in the formula is 2 or more. (Students may write WRONG chemical formulae such as $Ca(Cl)_2$, $(K)_2O$.) ✗ ✗

2 **Lithium oxide** The lithium ion (Li^+) carries 1 positive charge while the oxide ion (O^{2-}) carries 2 negative charges. For electrical neutrality, we need one O^{2-} ion for every two Li^+ ions. So the chemical formula of lithium oxide is Li_2O (Fig. 7.21). N12

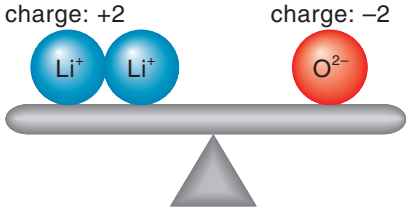


Fig. 7.21 Writing the chemical formula of lithium oxide

CE 2009 Paper 1 Q2(a)(ii)(1)

3 **Magnesium nitride** The magnesium ion (Mg^{2+}) carries 2 positive charges while the nitride ion (N^{3-}) carries 3 negative charges. For electrical neutrality, we need two N^{3-} ions for every three Mg^{2+} ions. So the chemical formula of magnesium nitride is Mg_3N_2 (Fig. 7.22).

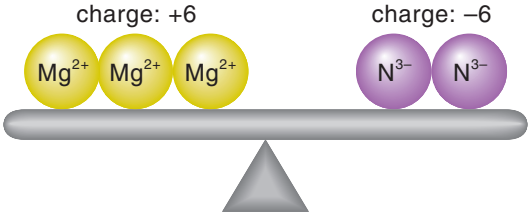


Fig. 7.22 Writing the chemical formula of magnesium nitride

4 **Calcium hydroxide** The calcium ion (Ca^{2+}) carries 2 positive charges while the hydroxide ion (OH^-) carries 1 negative charge. For electrical neutrality, we need two OH^- ions for every Ca^{2+} ion. So the chemical formula of calcium hydroxide is $Ca(OH)_2$ (Fig. 7.23).

If we do not write the brackets, the chemical formula would look like this: $CaOH_2$.

✘

This would read as 1 calcium, 1 oxygen and 2 hydrogens.

To make it simpler, we have omitted the charges on the ions here.

N13 Students may write WRONG chemical formulae such as $MgOH_2$, $CaNO_{32}$.

✘ ✘

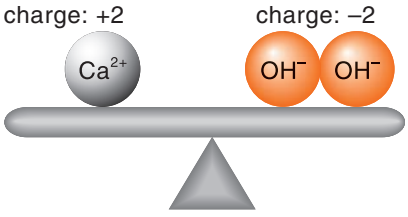


Fig. 7.23 Writing the chemical formula of calcium hydroxide

Notice that we use brackets for a polyatomic ion if the number of that ion in the chemical formula is 2 or more. N13

We can also use the following steps to work out the chemical formulae of ionic compounds. Let us take calcium carbonate, copper(II) nitrate and ammonium sulphate as examples (Table 7.8). N14

N14 Refer to page T30.

Table 7.8

Steps for working out the chemical formulae of ionic compounds

Step	Calcium carbonate	Copper(II) nitrate	Ammonium sulphate
1 Write down the symbols of ions in the compound.	Ca CO_3	Cu NO_3	NH_4 SO_4
2 Write down the number of charges of each ion on the top of each symbol.	2 2 Ca CO_3	2 1 Cu NO_3	1 2 NH_4 SO_4
3 Cross multiply the numbers and the symbols.	2 2 Ca CO_3 = Ca_2 = $(CO_3)_2$	2 1 Cu NO_3 = Cu = $(NO_3)_2$	1 2 NH_4 SO_4 = $(NH_4)_2$ = SO_4
4 Combine the symbols and simplify the ratio if necessary.	$CaCO_3$ (Simplify the ratio of 2 : 2 to 1 : 1.)	$Cu(NO_3)_2$	$(NH_4)_2SO_4$

Practice 7.3

1 Write down the names of the following compounds:

a) $MgCl_2$
magnesium chloride
b) Fe_2O_3
iron(III) oxide
c) $CuCO_3$
copper(II) carbonate

2 Work out the chemical formulae of the following compounds:

a) sodium sulphate
Na_2SO_4
b) lead(II) bromide
$PbBr_2$
c) ammonium carbonate
$(NH_4)_2CO_3$
d) calcium phosphate
$Ca_3(PO_4)_2$
e) iron(II) hydroxide
$Fe(OH)_2$
f) potassium dichromate
$K_2Cr_2O_7$

3 M is an element in the third period of the periodic table. It forms a hydroxide which has the chemical formula $M(OH)_3$. What is the chemical formula of the sulphate of M?
$M_2(SO_4)_3$; refer to page T33.

CE 2013 Paper 1A Q4

Animation
The structure of metals

7.10 Metallic bonds in metals

Metallic atoms tend to lose their outermost shell electrons. A metallic atom becomes a positively charged ion when it loses its outermost shell electron(s).

In a lump of solid sodium metal, the atoms are closely packed together. The outermost shell electron of each atom continually moves from one atom to another and is not associated with any specific atom. The mobile electron is said to be **delocalized**. The atom that the electron leaves behind becomes a positive ion. The delocalized electrons in the lump of sodium form a 'sea' of electrons that surrounds the regular three-dimensional arrangement of sodium ions (Fig. 7.24).

CE 2008 Paper 1 Q12(a)
AL 2005 Paper 1 Q1(a)(ii)
AL 2010 Paper 1 Q9

The lump of sodium is still made up of sodium atoms. The outermost shell electron of each sodium atom has not been lost. It just moves from one atom to another and does not attach to any particular atom. The electron is still there in the structure.

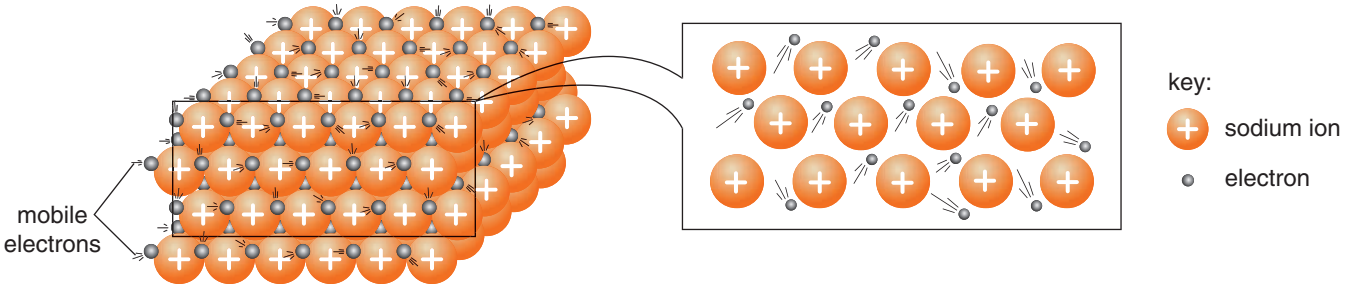


Fig. 7.24 A piece of sodium consists of sodium ions surrounded by a 'sea' of electrons N15

N15 Students should be able to describe the structure and bonding in metal with the help of a diagram.

N16 The binding forces in both metallic bond and ionic bond are non-directional.

Metallic bond differs from covalent bond in that metallic atoms do not form discrete molecules. However, the atomic cores are bound together by being attracted to the electrons between them in both metallic bond and covalent bond.

The attractive forces between the delocalized electrons and the positive metallic ions hold the structure together. This type of bonding is found only in metals and is called a **metallic bond**. N16

✓ A metallic bond refers to the strong electrostatic forces of attraction between the positive metallic ions and a 'sea' of delocalized electrons.

delocalized electron 離域電子　　metallic bond 金屬鍵

Have you mastered?

Key terms

conductor 導電體 75 electrolyte 電解質 75 molten 熔融的 75 non-conductor 非導電體 75
chemical bond 化學鍵 78 ionic bond 離子鍵 78 electrostatic force of attraction 靜電引力 79
ionic compound 離子化合物 79 polyatomic ion 多原子離子 83 hydroxide ion 氫氧離子 83
nitrate ion 硝酸根離子 83 carbonate ion 碳酸根離子 83 sulphate ion 硫酸根離子 83
ammonium ion 銨離子 83 permanganate ion 高錳酸根離子 85 dichromate ion 重鉻酸根離子 85
phosphate ion 磷酸根離子 85 chemical formula 化學式 89 delocalized electron 離域電子 92
metallic bond 金屬鍵 92

Checklist

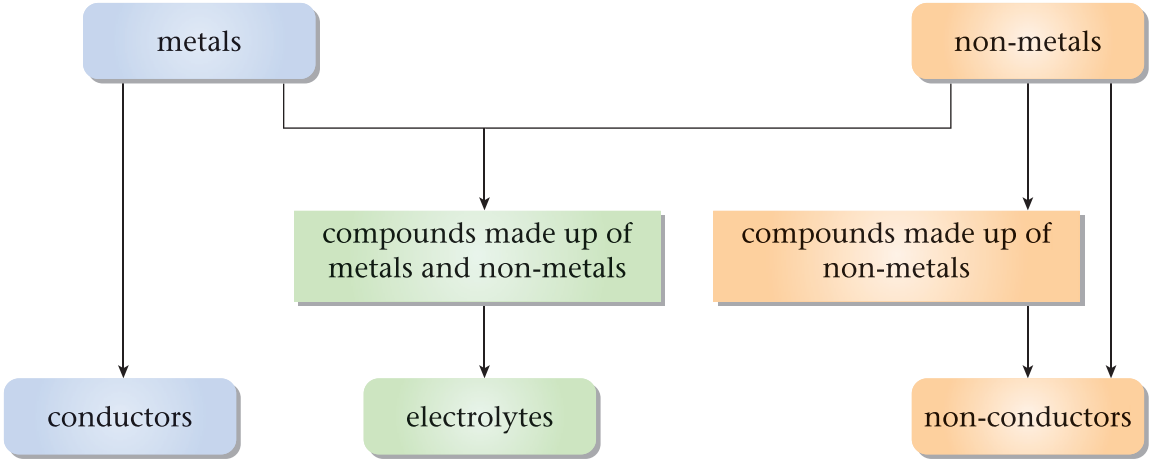
After studying this unit, you should be able to

- ☐ describe the formation of ionic bonds using electron diagrams;
- ☐ identify polyatomic ions;
- ☐ name some common cations and anions according to the chemical formula of ions;
- ☐ name ionic compounds based on the component ions;
- ☐ describe the colours of some common ions in aqueous solutions;
- ☐ interpret chemical formulae of ionic compounds in terms of the ions present and their ratios;
- ☐ construct chemical formulae of ionic compounds based on their names or component ions;
- ☐ describe the simple model of metallic bond.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Substances can be classified as conductors, electrolytes and non-conductors.

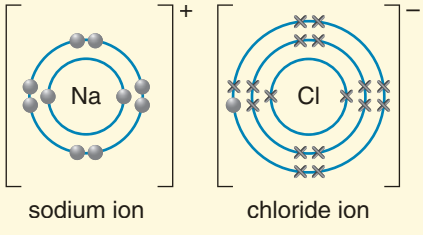
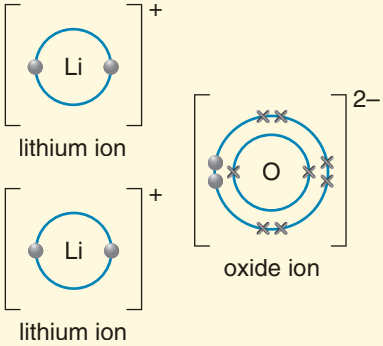


2 An ionic bond is the strong electrostatic forces of attraction between oppositely charged ions.

a) An ionic bond is formed when one or more electrons are transferred from one atom (or group of atoms) to another.

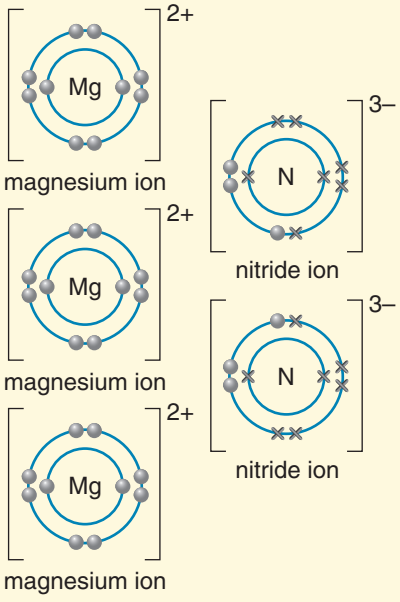
b) When a metal and a non-metal combine to form an ionic compound, the metallic atoms release electrons while the non-metallic atoms gain electrons.

3 The table below shows electron diagrams of some common ionic compounds.

Ionic compound	Electron diagram
Sodium chloride	 sodium ion chloride ion
Lithium oxide	 lithium ion lithium ion oxide ion

Continued on next page

Have you mastered?

Ionic compound	Electron diagram
Magnesium nitride	

4 A polyatomic ion is formed from a group of atoms, instead of a single atom.

5

Naming positive ions:	Examples
• If a metal forms only one kind of positive ion, the name of the ion is the same as the metal.	Na^+ (sodium ion) Mg^{2+} (magnesium ion)
• Some metals can form more than one kind of positive ion. Write a Roman numeral in brackets to show the number of positive charges.	Cu^+ (copper(I) ion) Cu^{2+} (copper (II) ion)

6

Naming negative ions:	Examples
• Simple negative ions have names ending in '-ide'.	Cl^- (chloride ion) O^{2-} (oxide ion)
• Polyatomic ions containing oxygen have names ending in -ite or -ate.	SO_3^{2-} (sulphite ion) SO_4^{2-} (sulphate ion)

7 When naming an ionic compound, name the positive ion first, followed by the negative ion.

Have you mastered?

8 If an ionic compound has colour, the colour may arise from either the negative or positive ion, or from both ions.

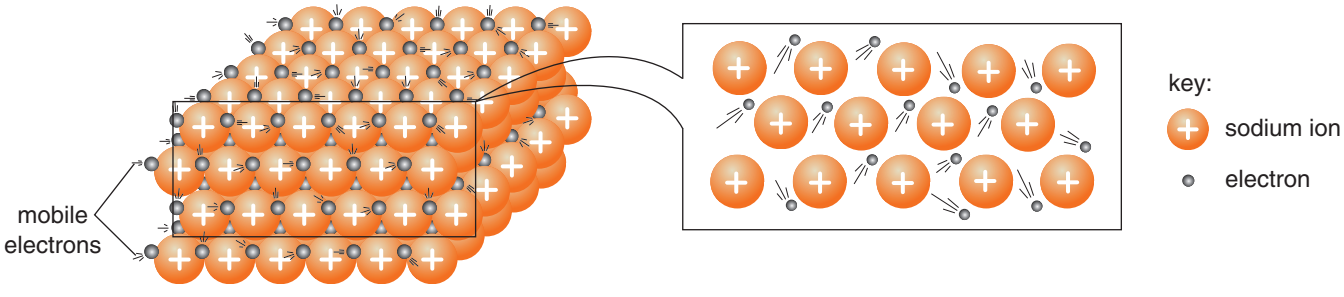
The following table lists the colours of some ions in aqueous solutions.

Ion	Chemical formula	Colour
Iron(II)	Fe^{2+}	pale green
Iron(III)	Fe^{3+}	yellow-brown
Copper(II)	Cu^{2+}	blue or green
Permanganate	MnO_4^-	purple
Dichromate	$Cr_2O_7^{2-}$	orange
Chromium(III)	Cr^{3+}	green
Nickel(II)	Ni^{2+}	green
Manganese(II)	Mn^{2+}	very pale pink (or colourless)

9 The chemical formula of an ionic compound shows the types of ions present and the ratio of one type of ion to the other.

10 A metallic bond refers to the strong electrostatic forces of attraction between the positive metallic ions and a 'sea' of delocalized electrons.

The following diagram shows the metallic bonding in a piece of metal.



Have you mastered?

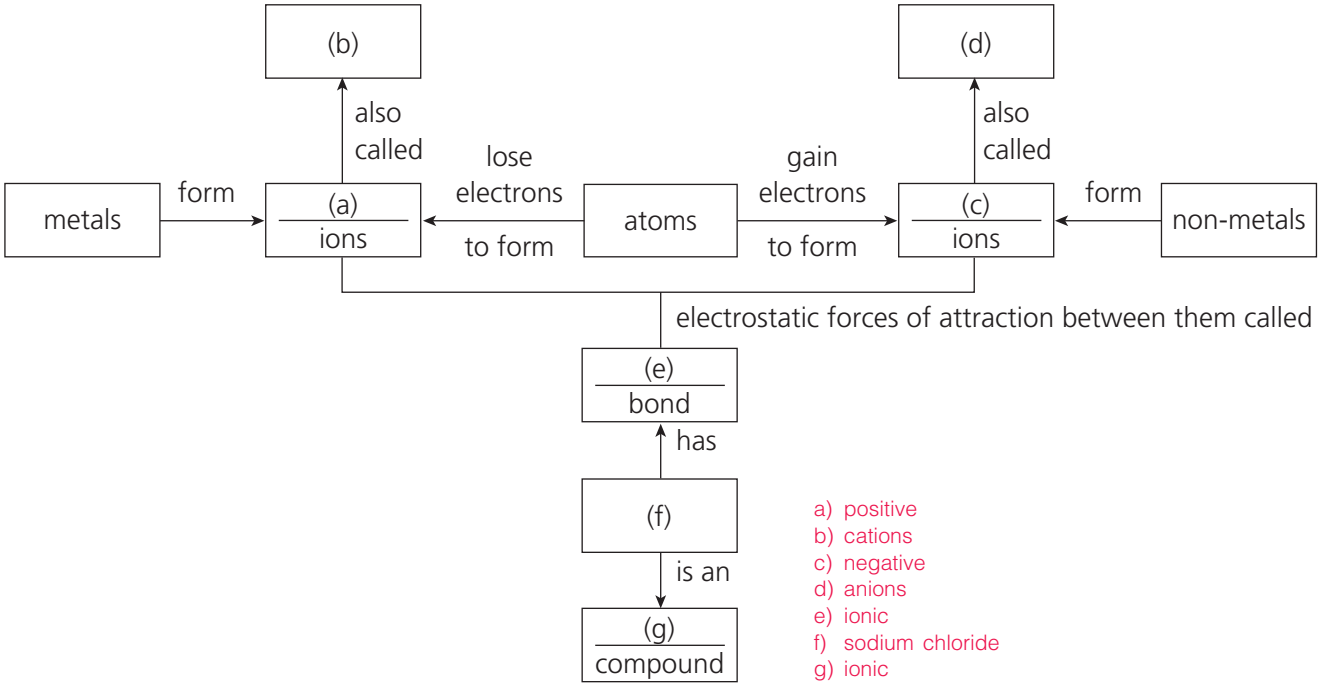
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the concept map using the following words:

anions, cations, ionic (2 times), negative, positive, sodium chloride



a) positive
b) cations
c) negative
d) anions
e) ionic
f) sodium chloride
g) ionic

2 Give the chemical formulae of the following compounds.

Name	Chemical formula
Lead(II) sulphate	$PbSO_4$
Iron(III) hydroxide	$Fe(OH)_3$
Calcium chloride	$CaCl_2$
Ammonium nitrate	NH_4NO_3
Aluminium oxide	Al_2O_3
Copper(II) carbonate	$CuCO_3$
Potassium hydrogencarbonate	$KHCO_3$
Sodium sulphite	Na_2SO_3

3 Give the names of the following compounds.

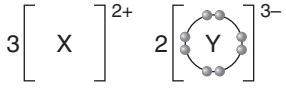
Chemical formula	Name
NH_4Cl	ammonium chloride
$Ca(NO_3)_2$	calcium nitrate
$FeSO_4$	iron(II) sulphate
PbO	lead(II) oxide
$ZnCO_3$	zinc carbonate
$FeCl_3$	iron(III) chloride
$Mg(OH)_2$	magnesium hydroxide
CaO	calcium oxide

4 Complete the following table on the colours of the ions in aqueous solutions.

Ion	Colour in aqueous solution
Iron(II)	pale green
Iron(III)	yellow-brown
Copper(II)	blue or green
Dichromate	orange
Permanganate	purple

Part II Multiple choice questions

5 Elements X and Y form a compound with the following electronic structure: Refer to page T33 for explanation.



(Only electrons in the *outermost shells* are shown.)

Which of the following combinations is correct?

	X	Y
A	Be	O
B	Al	S
(C)	Mg	N
D	C	O

6 Element R is in Group I of the periodic table and element T is in Group VI. R and T are NOT the symbols for the elements.

The compound of R and T will have the formula

A RT.
B RT_6.
C RT_2.
(D) R_2T.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2011, 7(a))

Element R is in Group I. Its atom tends to lose one electron to obtain a stable electronic arrangement.
Element T is in Group VI. Its atom tends to gain two electrons to obtain a stable electronic arrangement.
When R reacts with T, electrons from every two atoms of R, one from each, are transferred to one atom of T. Thus, the chemical formula of the compound is R_2T.

7 An atom of element X has 9 protons while an atom of element Y has 20 electrons. What is the chemical formula of the compound formed from X and Y?
Refer to page T33 for explanation.

A YX
B Y_2X
(C) YX_2
D Y_2X_3

8 M is an element in the third period of the periodic table. M forms a carbonate which has the chemical formula M_2CO_3. The chemical formula of the phosphate of M is likely to be

A M_2PO_4.
(B) M_3PO_4.
C $M_2(PO_4)_3$.
D $M_3(PO_4)_2$.

M is probably a Group I element. The ion of M (M^+) carries 1 positive charge. The phosphate ion (PO_4^{3-}) carries 3 negative charges. For electrical neutrality, we need three M^+ ions for every PO_4^{3-} ion. So the chemical formula of the phosphate of M is M_3PO_4.

9 Caesium (Cs) is a Group I element and its relative atomic mass is greater than that of potassium. Which of the following statements concerning caesium is INCORRECT?

A It is a soft metal.
B It reacts with water to give an alkaline solution.
C It reacts with chlorine to form an ionic compound.
(D) All of its compounds are coloured.

Compounds of Group I elements are usually colourless.

10 There is a gradual change in state as we move down Group VII of the periodic table. Iodine is a solid at room temperature and pressure. Thus, astatine should be a solid as well.

10 Astatine (At) is a halogen with an atomic number of 85. Which of the following statements concerning astatine is correct?

(A) Astatine is a solid at room temperature and pressure.
B Astatine is more reactive than chlorine.
C Astatine forms a salt with hydrogen.
D The chemical formula of the compound formed between astatine and potassium is K_2At.

11 Both strontium (Sr) and calcium (Ca) belong to the same group of the periodic table. Which of the following statements is INCORRECT?

A Both strontium and calcium become tarnished after having exposed to air for some time.
(B) Both strontium and calcium are stored in water.
C Strontium is more reactive than calcium.
D The chemical formula of strontium bromide is $SrBr_2$.

Strontium and calcium can react with water. They are NOT stored in water.

12 Which of the following statements best describes metallic bonding?

A It is an attractive force between ions.
B It is an attractive force between polar chemical species.
C It is an attractive force between atomic nuclei and bond-pair electrons.
(D) It is an attractive force between cations and delocalized electrons.

(HKDSE, Practice paper 1A, 2012, 15)

13 Element X (atomic number 11) combines with element Y (atomic number 16) to form a compound Z. Which of the following statements are correct?

Refer to page T34 for explanation.

(1) Z is an ionic compound.
(2) The chemical formula of Z is XY_2.
(3) When X combines with Y, each atom of X loses one electron.

A (1) and (2) only
(B) (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

14 The atomic number of element X is 20. Which of the following statements concerning X are correct?

(1) X can combine with oxygen to form an ionic compound.
(2) X is in Group II of the periodic table.
(3) The carbonate of X is soluble in water.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

Element X is calcium.
(3) Calcium carbonate is insoluble in water.

15 X, Y and Z are three consecutive elements in the periodic table. X forms a stable anion X^-, while Z forms a stable cation Z^+. Which of the following statements about X, Y and Z is / are correct?

(1) X^- and Z^+ have the same electronic arrangement.
(2) Y and Z are in the same period of the periodic table.
(3) X reacts with Y readily.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

16 Which of the following ions is / are green in colour?

(1) $Cr^{3+}(aq)$ ion
(2) $Fe^{3+}(aq)$ ion
(3) $Ni^{2+}(aq)$ ion

A (1) only
B (2) only
(C) (1) and (3) only
D (2) and (3) only

(2) $Fe^{3+}(aq)$ ion is yellow-brown in colour.

15 X is a Group VII element while Z is a Group I element.
Y is a noble gas.
(1) electronic arrangement of X^- = electronic arrangement of Y = electronic arrangement of Z^+
(2) Y and Z belong to different periods of the periodic table.

Part III Structured questions

Answers for the questions in Hong Kong public examinations are not provided.

17 Sodium and chlorine react to form sodium chloride.

a) A sodium atom has a mass number of 23 and an atomic number of 11.

Use this information to complete the table to show the number of protons, neutrons and electrons in this atom.

The information for a chlorine atom has been provided below.

Atom	Number of protons	Number of neutrons	Number of electrons
Chlorine	17	18	17
Sodium	11	12	11

b) i) Describe, in terms of electron transfer, how a sodium atom and a chlorine atom react to form particles in sodium chloride. One electron is transferred from one sodium atom to one chlorine atom.

ii) Give the name of the type of bonding in sodium chloride. Ionic

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Jun. 2009, 2(a)–(b)(ii))

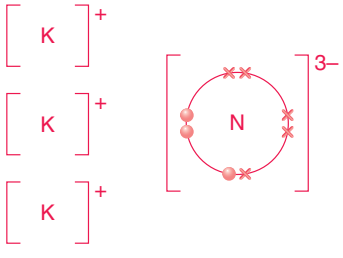
18 The elements in Group I of the modern periodic table show similar chemical properties.

Li
Na
K
Rb

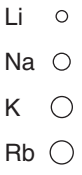
a) What is the name given to Group I elements of the periodic table?
Alkali metals

b) State the trend in the melting point of elements going down Group I.
Decrease

18 d)



c) The diagram below shows the relative atomic size of the Group I elements.



The number of occupied electron shells in atom of Group I elements increases with atomic number.

Explain why the relative atomic size of Group I elements increases with atomic number.

d) Potassium reacts with nitrogen to form compound P. Draw an electron diagram of P (showing electrons in the *outermost shells* only).

e) Give the chemical formula of the sulphate of rubidium (Rb).
Rb_2SO_4

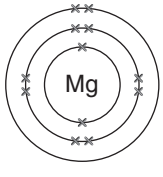
19 In the late 1800s, photographers burned magnesium to produce a magnesium flashlight.

This was used for indoor photography.



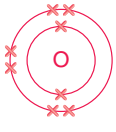
a) The atomic number of magnesium is 12.

The diagram below shows the electronic structure of a magnesium atom.



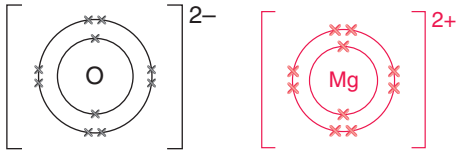
The atomic number of oxygen is 8.

Draw a similar diagram to show the electronic structure of an oxygen atom.



b) When magnesium reacts with oxygen, magnesium ions and oxide ions are made.

The diagram below shows the electronic structure of an oxide ion.



Draw a similar diagram to show the electronic structure of a magnesium ion.

c) Sodium oxide is another ionic compound.

Sodium ions, Na^+, combine with oxide ions, O^{2-}, to form sodium oxide.

Write down the chemical formula for sodium oxide. Na_2O

(OCR GCSE Gateway Science, Chem. B, Unit 1, Jan. 2012, 13(b)–(c), (e))

20 In the periodic table, the element fluorine is shown as

19
F
fluorine
9

a) All fluorine atoms are identical.

State the name, number and position of the particles in one atom of fluorine.
9 protons and 10 neutrons in nucleus; 9 electrons in shells

b) Calcium fluoride is ionic. It contains calcium ions, Ca^{2+}, and fluoride ions, F^-.

i) What is an ion? An ion is formed when an atom either loses or gains electron(s).

ii) What is the chemical formula of calcium fluoride? CaF_2

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Mar. 2009, 2(a)–(b)(ii))

21 Aluminium oxide contains aluminium ions and oxide ions.

a) Draw an electron diagram of aluminium oxide, showing electrons in the *outermost shells* only.

b) State the chemical formula of aluminium oxide.
Al_2O_3

c) What causes the aluminium ions and oxide ions to be held together strongly? Electrostatic forces of attraction between oppositely charged ions.

22 The following table shows some information about the atoms of four elements. Refer to page T34.

Element	Number of protons in atom	Number of neutrons in atom
W	3	4
X	7	7
Y	16	16
Z	19	20

a) Which of the above elements belong to the same group? Explain your answer.

b) W and X combine to form an ionic compound P.

i) Draw an electron diagram of P, showing electrons in the *outermost shells* only.

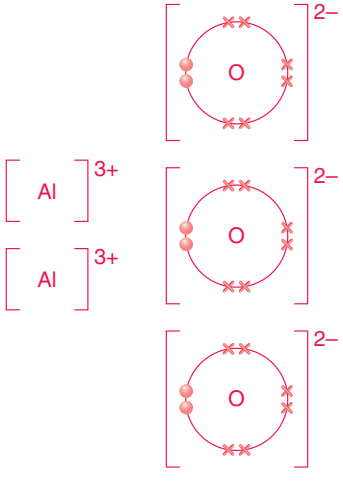
ii) State the chemical formula of P.

c) Y and Z combine to form an ionic compound Q.

i) Draw an electron diagram of Q, showing electrons in the *outermost shells* only.

ii) State the chemical formula of Q.

21 a)



23 Magnesium can burn in air under strong heating.

a) State the expected observation when magnesium burns in air.

b) Magnesium nitride is also formed when magnesium burns in air.

i) State the chemical formula of magnesium nitride.

ii) Draw the electron diagram of magnesium nitride, showing electrons in the *outermost shells* only.

(HKCEE, Paper 1, 2009, 2(a))

24 In July 2009, the element with atomic number 112 was named copernicium. The element was named after Nicolaus Copernicus, a Polish astronomer.



Nicolaus Copernicus

The symbol for an atom of copernicium is Cp.

a) The atomic number of copernicium is 112.

i) How many protons are there in an atom of copernicium?
112

ii) Some atoms of copernicium have a mass number of 277.

How many neutrons are in the nucleus of one of these atoms?
165

iii) By July 2009, four isotopes of copernicium were known.

How do atoms of these isotopes of copernicium differ from each other?
They have different number of neutrons.

b) ii) When a copernicium atom loses two electrons, a copernicium ion forms.
When a fluorine atom gains one electron, a fluoride ion forms.

b) As only a few atoms of copernicium have been made, its chemical and physical properties are unknown.

However, copernicium is expected to form an ionic fluoride.

This compound will contain copernicium ions, Cp^{2+}, and fluoride ions, F^-.

i) Write the chemical formula of copernicium fluoride.
CpF_2

ii) Describe how copernicium atoms, Cp, and fluorine atoms, F, form copernicium ions, Cp^{2+}, and fluoride ions, F^-.

(Edexcel GCSE (Higher Tier), Chemistry, Unit C2, Nov. 2010, 2)

25 Strontium (Sr) and calcium (Ca) are elements in Group II of the periodic table. Refer to page T34.

a) What is the name commonly given to this group of elements?

b) i) The atomic number of strontium is 38. Write the electronic arrangement of a strontium atom.

ii) State the similarity between a calcium atom and a strontium atom in terms of electronic arrangement.

c) Strontium reacts with nitrogen to form an ionic compound.

i) Draw an electron diagram of the compound, showing electrons in the *outermost shells* only.

ii) Give the name of the compound.

d) i) What is the chemical formula of strontium phosphate?

ii) How many elements does strontium phosphate consist of?

e) A sample of strontium contains three isotopes: ^{86}Sr, ^{87}Sr and ^{88}Sr. The relative atomic mass of this sample is 87.73. The relative abundance of the ^{88}Sr isotope is 83.1%. What is the relative abundance of the ^{86}Sr?
10.1%

26 Elements can be broadly classified as 'metals' and 'non-metals'.

a) Give TWO typical PHYSICAL properties of zinc and TWO typical PHYSICAL properties of sulphur to illustrate the difference between a metal and a non-metal.

b) i) Magnesium atom has an electronic arrangement of 2,8,2 and oxygen atom has an electronic arrangement of 2,6. Explain in terms of electron transfer how magnesium oxide is formed.

ii) Explain why two metals do not combine to form a compound.

c) State which group of non-metals in the periodic table is unreactive and explain why this is so.

(Edexcel GCE O Level, Paper 2, Jun. 2010, 8(a)(i), b(i), (iii), (d))

27 This question compares the bonding of sodium and sodium oxide. Refer to page T35.

a) Sodium, Na, is a metallic element.

Explain, with the aid of a labelled diagram, what is meant by the term 'metallic bonding'.

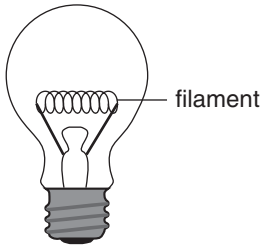
b) Sodium reacts with oxygen to form sodium oxide, Na_2O, which is an ionic compound.

i) State what is meant by the term 'ionic bond'.

ii) Draw an electron diagram to show the bonding in Na_2O, showing electrons in the *outermost shells* only.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2009, 2(a), b(ii)–(iii))

28 The simplified diagram below shows a tungsten light bulb. The filament inside is made of tungsten metal (atomic number = 74). The filament can emit light by passing an electric current until it reaches an extremely high temperature. Under vacuum condition, the filament would emit tungsten particles under extremely high temperature and blacken the light bulb. To avoid this from happening, the light bulb should be filled up with gas.



a) Describe the structure and bonding in tungsten metal with the help of a diagram.

b) Apart from the property that tungsten can emit light at extremely high temperature, suggest another PHYSICAL property that can render it suitable for making filament. Explain why it has this property.

c) Tungsten is stable in air under normal conditions. However, it is not appropriate to fill up a tungsten light bulb with air. Explain.

d) Suggest, with explanation in terms of electronic arrangement, a suitable gas for filling up a tungsten light bulb.

e) Explain, from the perspective of energy, why the efficiency of tungsten light bulb is not high enough.

(HKCEE, Paper 1, 2008, 12)

26 a) Zinc
Any two of the following:
- Malleable
- Ductile
- High melting point / boiling point
- Shiny
- Good conductor of electricity / heat

Sulphur
Any two of the following:
- Brittle
- Low melting point / boiling point
- Dull
- Non-conductor of electricity

b) i) When magnesium reacts with oxygen, two electrons from each magnesium atom are transferred to one oxygen atom. Each atom becomes an ion with an electronic arrangement the same as that of a neon atom.
Mg^{2+} ions and O^{2-} ions are formed.
An ionic compound, magnesium oxide, is formed.
ii) Both metals need to lose electrons.

c) Noble gases
Their atoms have stable electronic arrangements.